

Equivalence Partitioning Guide

Slash your test count without losing coverage.

Introduction

Equivalence Partitioning (EP) is a black-box technique that divides the input (or output) data into partitions where the system is expected to behave the same way. You then test just one representative value from each partition, on the principle that if it works for one, it works for all. This guide shows you how to partition, a worked example and how EP pairs with Boundary Value Analysis.

The Core Idea

If every value in a group should be treated identically by the system, testing one value from that group is as good as testing them all. This dramatically reduces the number of test cases while keeping systematic coverage. Partitions can be valid (accepted values) or invalid (rejected values).

Valid vs Invalid Partitions

Partition type	Purpose	Expected result
Valid	Values the system should accept and process	Accepted / processed correctly
Invalid	Values the system should reject	Handled gracefully (error message, rejection)

Worked Example — age field valid for 18 to 65

The specification says an applicant must be aged 18 to 65 inclusive. That gives three partitions: too young, valid, too old. Pick one representative from each.

Partition	Range	Representative value	Expected result
Invalid — too young	0 to 17	10	Rejected
Valid	18 to 65	40	Accepted
Invalid — too old	66 and above	70	Rejected

Three test cases (10, 40, 70) give full equivalence-partition coverage of this field. Note that EP picks values from the middle of each partition; the edges (17, 18, 65, 66) are the job of Boundary Value Analysis.

Step-by-Step Method

1. Identify the input or output under test and its specification.
2. Divide the data into partitions where behaviour should be identical.
3. Include both valid and invalid partitions.
4. Choose one representative value from each partition (typically a mid-range value).
5. Design one test case per partition, recording the expected result.

Coverage & Combining with BVA

Minimum EP coverage is one test per partition. In practice you almost always combine EP with Boundary Value Analysis: EP tells you the partitions, BVA tests their edges. Together they give strong coverage of a numeric field for very few cases.

Exam Traps & Common Mistakes

- Forgetting the invalid partitions — testers often only pick valid values.
- Choosing boundary values as representatives; EP wants a mid-partition value, BVA handles the edges.
- Assuming partitions never overlap — a well-formed partition set is mutually exclusive and covers all possible values.
- Partitioning only inputs — outputs and even internal data can be partitioned too.

COVERAGE TIP

EP coverage is measured as the number of partitions covered divided by the total number of partitions. Cover every valid and invalid partition at least once to reach 100%.

Best Practices

- Always partition both valid and invalid data.
- Use EP to decide "which values", then BVA for "which edges".
- Document each partition and its representative so coverage is auditable.

INSIDE STLC PRO TIP

EP and BVA are the classic pairing in the ISTQB exam and in real test design. Whenever you see a numeric or range-based field, reach for both: EP for the partitions, BVA for the boundaries.